RAS545: Final Project
Smart and Accurate Path Planning with the MyCobot Pro 600: Solving Mazes

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ABSTRACT (10 PT)

This report presents the development and evaluation of a smart and accurate maze-solving path planning system for the MyCobot Pro 600. Building on prior work in straight and curved path planning (Labs 3 and 4), we implement a fully automated pipeline that captures a maze image, computes the shortest solution path, maps pixel coordinates to physical workspace coordinates, performs inverse kinematics via a digital twin, and finally drives the physical robot to traverse the maze within quick succession.

1. INTRODUCTION

This lab follows the previous linear path planning and curved trajectory path planning lab work. Extracting object coordinates via OpenCV and validating movements in a Simulink digital twin before commanding the physical robot to follow the coordinates and perform inverse kinematics. This final project elevates the previously execute lab work by implementing an intelligent maze solver. From maze detection through an image to shortest path computation and time efficient physical execution.

2. METHODOLOGY

Maze Image Acquisition and Processing

A camera captures the 15 cm×15 cm maze workpiece. Using OpenCV's adaptive thresholding and morphological thinning, the binary maze walls were isolated. Start (red circle) and end (green dot) points were detected via color segmentation in HSV space. A python code was written to integrate all of this.

Maze Solving Algorithm

The binary maze image was converted into a grid graph where free pixels map to nodes. We applied A* search with Manhattan distance as the heuristic to compute the shortest path from start to end. This choice balances optimality with computational efficiency for

mazes up to 100×100 pixels. This would give us the most optimal path or coordinates of the input maze image.

Pixel-to-Physical Coordinate Mapping

Calibration was done then 10 known markers were plotted in the image to their real-world counterparts. A matrix was also computed to map each path pixel (u,v) to physical coordinates (x,y) in the robot's workspace. A fixed z-height of 50 mm above the workpiece was maintained for all waypoints.

Digital Twin and Inverse Kinematics

The digital twin developed in previous labs was reused to verify feasibility of each coordinate given to it. Inverse Kinematics solved for joint angles, ensuring collision avoidance by checking joint limits and inter-link distances. Simulation confirmed smooth, collision-free motion through all waypoints within tolerances of some degree. This part was for us to test whether the robot would move according to the coordinates provided by the script.

Robot Execution

Joint angle commands were sent via TCP set_angles() function command with a 0.5 s wait using wait_command_done() function between each move. The robot's home position was set outside the camera view with downward end-effector orientation to avoid occlusions. The robot followed the given coordinates and was able to smoothly get through the maze without hitting any of the walls.

3. RESULTS

- Path Planning Accuracy: A* reliably found the 32-segment shortest path.
- Simulation Validation: Digital twin executed the full path in 28 s, with Cartesian endpoint error ≤1.8 mm.
- Physical Execution: The MyCobot Pro 600 completed the maze traversal in a short amount of time without collisions or wall contacts.
- Efficiency: Total path length was 450 mm, matching the optimal solution identified in simulation.
- Robustness: The system handled three different maze samples with varying complexity, maintaining timing under 60 seconds in all trials.

These results demonstrate that integrating a graph-based solver with calibrated coordinate mapping and a validated digital twin yields both speed and accuracy in complex path planning tasks

4. CONCLUSION

The final project successfully extends linear and curved path planning to autonomous maze solving. By leveraging computer vision, A* search, coordinate transformations, and a proven digital twin platform, the MyCobot Pro 600 autonomously navigates mazes in

quick succession with millimeter-level precision. This framework provides a scalable basis for future vision-guided robotic applications requiring intelligent decision making and real-time execution. Overall, it was able to complete the maze smoothly moving in between the walls.